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Anatomical and Histochemical Fingerprint of Gastrointestinal Tract in Bony Fishes from Different Trophic Levels

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Abstract: Anatomy, morphology, and histology of the gastrointestinal tract (GIT) show ontogeny and phylogeny variations that are an evolutionary adaptation to feeding habits. The GIT includes the headgut (mouth and pharynx), foregut (esophagus, stomach, and pyloric ceca), midgut (intestine), and hindgut (rectum) in fish. It has extensive endocrine functions and plays important roles in food uptake and digestion, and nutrient uptake. In the present study, we aimed at an anatomical and histochemical fingerprint of the GIT in fish from different trophic levels. The GIT were sampled from omnivorous (tench, *Tinca tinca*, and common bream, *Abramis brama*) and carnivorous (donkey croaker, *Pennahia anea*) fishes. The GIT was sectioned and embedded in paraffin after dehydration. Serial sections of 7 µm thickness were stained with H&E or PAS. The histological slides were observed under light microscopy, and images were captured for histochemical characterization and GIT morphometry. Unlike *T. tinca*, a stomach was present in *A. brama* and *P. anea*. Pyloric ceca were not observed in any species. For all species, the GIT was composed of mucosa, submucosa, muscularis, and serosa layers. Morphometry of mucosal folds and cellular components varied along the GIT and showed species variations. Gastric glands and goblet cells were common components of the stomach and postgastric region, respectively; higher numbers were observed at the anterior stomach and posterior intestine. Histochemical traits were tightly aligned with feeding behavior. This study quantifies histological features of the GIT, illustrates the spatial distribution of cellular components along the GIT, provides a framework for understanding trophic specialization, and expands insight into the evolution of the digestive system.

Keywords: Gastric cells, Goblet cells, Intestine, Mucosal folds, Stomach